

Hotel Booking Demand: Data Quality Report

1. Automated Quality Check Log

The following sequence represents the real-time execution of the data validation scripts.

Data Quality Check Start...

Feature	Validation Status	Observed Issue / Result
adr ▾	FAILED :(▾	Initial check detected values ≤ 0 , representing comped rooms or data entry errors.
hotel ▾	FAILED :(▾	Null values detected; this is a mandatory categorical field for hotel identification.
lead_time ▾	FAILED :(▾	Detected negative values ($\text{lead_time} < 0$) which violate temporal business logic.
adr ▾	PASS :) ▾	Re-validated after filtering for values > 0 and applying outlier capping/Winsorization.
adults ▾	FAILED :(▾	Violation detected: Adults < 0 or total guest count ($\text{Adults} + \text{Children} + \text{Babies}$) = 0.
Total Guests ▾	PASS :) ▾	Engineered feature successfully calculated as the sum of Adults, Children, and Babies.

2. Detailed Failure Analysis & Business Impact

The failures identified above represent critical risks to the accuracy of the revenue forecasting and booking behavior models.

Financial Integrity (ADR)

- **Rule:** ADR must be greater than 0.
- **Impact:** Records with an ADR of 0 represent "comped" or free rooms. If not filtered, these skew average revenue metrics and pricing models.
- **Status:** Successfully resolved in the second pass through data cleaning and filtering.

Structural Completeness (Hotel & Lead Time)

- **Hotel Rule:** Field should not be null.
- **Lead Time Rule:** Must be ≥ 0 .
- **Impact:** Missing hotel identifiers prevent location-based performance comparisons (City vs. Resort). Negative lead times are logically impossible and indicate data corruption.

Operational Validation (Adults & Total Guests)

- **Rule:** Adults + Children + Babies > 0 .
- **Impact:** A booking cannot exist without at least one guest. These "zero-guest" records are typically data entry errors that must be imputed or removed to prevent skewed occupancy reports.

3. Data Remediation Recommendations

Based on the EDA Report and Machine Learning Readiness guidelines, the following steps are required to move to the analysis phase:

- **Missing Value Treatment:** Drop features with extreme missing data, such as `company` (94%) and `agent` (13%), and impute `children` and `country`.
- **Duplicate Removal:** Remove the 31,994 duplicate records to prevent skewed revenue forecasting.
- **Scaling:** Apply `RobustScaler` to numerical features like `lead_time` to handle the heavy right-skew and outliers identified during initial checks.

- **Data Leakage Prevention:** Ensure that `reservation_status` and `reservation_status_date` are excluded from predictive models, as they represent "future knowledge".